Algorithmic Trading: model reference

Arthur Li

October 25, 2026

Contents

Notation and conventions ii

Model index iii

1 Market microstructure and market making 1

1.1 Market making, queues and limit-order-book models . . . 1

1.1.1 Grossman–Miller market making . . . 1

1.1.2 Kyle single-auction model . . . 2

1.1.3 Static limit-order depth . . . 3

1.1.4 Roll's serial-covariance spread estimator . . . 3

Sources outside the catalogue 5

Notation and conventions

Notation follows the compendium entries each derivation serves; the compendium's general notation list applies otherwise. Prices are in currency per unit, quantities in units (shares), and variances of one period's shock in squared price units, not annualised. The symbols below are local to a model, and some are reused with different meanings across models.

Symbol	Meaning	Model
$t \in \{1, 2, 3\}$	dates: first liquidity trade, arrival of the offsetting trader, settlement (date 0 added for the reversal)	Grossman–Miller
S_t, μ, ϵ_t	date-t trade price, expected terminal value, value shock announced before date t	Grossman–Miller
q_t^j, Q_1^{LT1}	agent j's holding on leaving date t; LT1's date-1 sale (a quantity traded, not a holding)	Grossman–Miller
n, i, γ, σ^2	market makers, LT1's initial holding, absolute risk aversion, one-period shock variance	Grossman–Miller
η, p	per-share fee paid by each side; liquidity discount $\lvert\mu - S_1\rvert$	Grossman–Miller
$\lambda = -\gamma\sigma^2/n$	slope of S_1 on the quantity LT1 sells; negative	Grossman–Miller
v, u, x, y	asset value, noise order, insider's order, net order flow	Kyle
λ, β	price-impact slope on net order flow (positive, price per share); insider's trading intensity	Kyle
$\delta^{\pm}$, $\kappa^{\pm}$, $p^{\pm}$	depth of the sell (buy) order from S_t; fill-decay rate; probability of a buy (sell) market order, not a price	static limit-order depth
P_t, V_t, Q_t, ε_t	trade price, efficient price, trade sign ± 1, efficient-price innovation of variance σ^2	Roll
S	the full bid–ask spread, in price units, not a price	Roll

Each derivation states the assumptions under which its results hold and names its sources with locators. The working is own derivation following the cited textbook accounts; it records source checking, not study. The closed forms are checked numerically, with fixed parameters, seeds and tolerances, by `trading_notes.numerical.market_making_models.lob_model_derivations_checks`.

Model index

Each row maps a result to its compendium entry (a link into `compendium/compendium.pdf`), its derivation in this reference, its implementation in `trading_notes.numerical.market_making_models.microstructure_models`, and the part of `trading_notes.numerical.market_making_models.lob_model_derivations_checks` that checks it. The checks module runs from any directory, prints one line per comparison and exits with status 1 on a failure.

Model	Compendium entry	Derivation	Implementation and numerical check
Grossman–Miller market making	`mod:grossman-miller`	`drv:grossman-miller`	`GrossmanMillerFees` with $\eta = 0$; the date-1 price and holdings are those of part 1 at zero fee and are not checked separately
Per-share trading fees	`res:grossman-miller-trading-fees`	`drv:grossman-miller-trading-fees`	`GrossmanMillerFees.price`, `.trader_demand`; part 1: the date-1 price and LT1's holding by root-finding (`brentq`, `xtol` 10^{-14}) against the closed forms at $\mu = 10$, $\gamma = 2$, $\sigma^2 = 0.04$, $n = 5$, $i = 3$, $\eta = 0.01$; tolerance 10^{-9}
Liquidity measures	`res:grossman-miller-liquidity-measures`	`drv:grossman-miller-liquidity-measures`	`GrossmanMillerFees.liquidity`; part 2: $\lambda = -\gamma\sigma^2/n = -0.016$, and $\mathrm{Cov}(\Delta_1, \Delta_2) = -\lambda^2 \mathrm{Var}(Q)$ by simulation (seed 1, $N = 400{,}000$, $Q \sim N(0,1)$, news $\epsilon_1, \epsilon_2 \sim N(0, 0.1^2)$); tolerance 3×10^{-5}
Kyle single auction	`mod:kyle-single-auction`	`drv:kyle-single-auction`	`KyleAuction`; part 3: $\beta, \lambda = 1/3$, expected profit 3 and $\mathrm{Var}(v \mid y) = 2$ at $\sigma_v = 2$, $\sigma_u = 3$, mean 1, by simulation (seed 2, $N = 1{,}000{,}000$), relative tolerance 1 percent; the projection slope by regression, tolerance 0.005
Static limit-order depth	`mod:static-limit-order-depth`	`drv:static-limit-order-depth`	no domain class; part 4: $\delta^\star = 1/\kappa$ maximises $\delta e^{-\kappa\delta}$ on a grid of 200,001 points on $[0, 2]$ with $\kappa = 4$, tolerance 10^{-5}; second-order condition negative
Roll spread estimator	`res:roll-spread-estimator`	`drv:roll-spread-estimator`	`RollModel`; part 5: $\mathrm{Var}(\Delta P) = \sigma^2 + S^2/2$ and $\mathrm{Cov}(\Delta P_t, \Delta P_{t-1}) = -S^2/4$ at $S = 0.2$, $\sigma = 0.1$, by simulation (seed 3, $N = 1{,}000{,}000$), tolerance 5×10^{-4}; lag-2 autocovariance within 5×10^{-4} of zero

The first five entries are in the compendium chapter *Market microstructure and market making*, pathway *Market making, queues and limit-order-book models*, topic file `execution-market-making/market-making-models.tex`; Roll's estimator is in the same chapter's *Market organisation, informed trading and liquidity measurement* pathway, topic file `execution-market-making/spread-measurement.tex`. The derivations follow the original grouping of the four models in one topic, `execution-market-making/market-making-models/`.

1 Market microstructure and market making

1.1 Market making, queues and limit-order-book models

1.1.1 Grossman–Miller market making

Purpose. Derive the prices, holdings and trades of the three-date Grossman–Miller market, the version with per-share fees, and the two liquidity measures it implies. Compendium entries: `mod:grossman-miller`, `res:grossman-miller-trading-fees`, `res:grossman-miller-liquidity-measures`.

Sources. [`crypto-cartea`, chap. "A Primer on the Microstructure of Financial Markets", pp. 21–27]; the free-entry condition from [`grossman-miller-1988-nber`, eqs. (16)–(19), printed pp. 18–19].

Implementation and checks. `trading_notes.numerical.market_making_models.microstructure_models.GrossmanMillerFees`; `trading_notes.numerical.market_making_models.lob_model_derivations_checks` parts 1 and 2 (model index).

Assumptions

i. Dates $t \in \{1, 2, 3\}$: date 1 is the first liquidity trade, date 2 the arrival of the offsetting liquidity trader, date 3 terminal cash settlement.
ii. Cash earns nothing; the risky asset is worth $S_3 = \mu + \epsilon_2 + \epsilon_3$ at date 3, with ϵ_2, ϵ_3 independent $N(0, \sigma^2)$ and ϵ_t announced between dates $t-1$ and t.
iii. n identical market makers (MM) start with no inventory; liquidity trader LT1 holds i units and LT2, present only from date 2, holds $-i$. For $i > 0$ LT1 is a natural seller and LT2 a later buyer.
iv. Every agent has CARA utility $U(X) = -\exp(-\gamma X)$, $\gamma > 0$.
v. No trading, funding or holding costs, except where per-share fees are introduced below; the only cost of supplying liquidity is carrying inventory between dates 1 and 2. LT2's arrival at date 2 is certain.

Variables and timing. Let q_t^j be agent j's holding on leaving date t and S_t the date-t trade price. Prices are in currency per unit and holdings in units; σ^2 is the variance of one period's value shock, in squared price units and not annualised, and γ is in inverse currency units. At date t every present agent knows the shocks announced before it. Wealth evolves as

$$X_2^j = X_1^j + q_1^j (S_2 - S_1), \qquad X_3^j = X_2^j + q_2^j (S_3 - S_2).$$

The CARA–normal demand

If terminal wealth is $X + q(Z - S)$ with $Z \sim N(m, s^2)$ given the agent's information, then

$$\mathbb{E}\big[U(X + q(Z - S))\big] = -\exp\Big(-\gamma\big(X + q(m - S)\big) + \tfrac{1}{2}\gamma^2 q^2 s^2 \Big),$$

by the normal moment-generating function. Maximising expected utility is therefore maximising the certainty equivalent $X + q(m - S) - \frac{1}{2}\gamma q^2 s^2$, which is strictly concave in q; the first-order condition $m - S - \gamma q s^2 = 0$ gives the demand

$$q^\star = \frac{m - S}{\gamma s^2}.$$

Date 2

Given ϵ_2, the next value is S_3 with $m = \mathbb{E}[S_3 \mid \epsilon_2] = \mu + \epsilon_2$ and $s^2 = \sigma^2$, so every agent present demands

$$q_2^j = \frac{\mathbb{E}[S_3 \mid \epsilon_2] - S_2}{\gamma \sigma^2}.$$

All $n + 2$ agents are present. Holdings entering date 2 sum to $n q_1^{MM} + q_1^{LT1} + q_1^{LT2} = i - i = 0$, because trading at date 1 only redistributes LT1's i units and LT2 enters with $-i$. Market clearing requires the date-2 holdings to sum to the same total:

$$0 = (n + 2)\,\frac{\mathbb{E}[S_3 \mid \epsilon_2] - S_2}{\gamma \sigma^2} \quad \Longrightarrow \quad S_2 = \mathbb{E}[S_3 \mid \epsilon_2] = \mu + \epsilon_2, \qquad q_2^j = 0.$$

Date 2 is the unwind: the matching liquidity has arrived and no agent wants risky inventory.

Date 1

At date 1, LT2 is absent. Each present agent anticipates the date-2 price $S_2 = \mu + \epsilon_2$, so $m = \mathbb{E}[S_2] = \mu$, $s^2 = \mathrm{Var}(S_2) = \sigma^2$, and

$$q_1^j = \frac{\mu - S_1}{\gamma \sigma^2}, \qquad j \in \{MM, LT1\}.$$

Clearing among the n market makers and LT1, whose holdings entering date 1 are 0 and i:

$$i = n q_1^{MM} + q_1^{LT1} = (n + 1)\,\frac{\mu - S_1}{\gamma \sigma^2},$$

so

$$q_1^{MM} = q_1^{LT1} = \frac{i}{n + 1}, \qquad S_1 = \mu - \gamma \sigma^2 \,\frac{i}{n + 1}.$$

Result and interpretation. LT1 sells $i - \frac{i}{n+1} = \frac{n}{n+1}\, i$ at date 1 and each market maker holds $\frac{i}{n+1}$ until LT2 arrives. The liquidity discount is $p = |\mu - S_1| = \gamma \sigma^2 \, |i| \,/ (n+1)$; for $i < 0$ the same argument gives a premium. Interpretation: a bid at $\mu - p$ and an ask at $\mu + p$ imply an inside spread of $2p$.

The concession is a price for risk, not information: no agent is informed and the imbalance nets to zero across dates. It rises with $|i|$, γ and σ^2 and falls with n; as $n \to \infty$, $S_1 \to \mu$ and LT1's immediate trade tends to i.

Free entry. In the original, a market maker forgoes opportunities worth c, and entry occurs until $\mathbb{E}U(W_0 - c + (\tilde{P}_2 - P_1)x_1^m) = \mathbb{E}U(W_0)$ [grossman-miller-1988-nber, eq. (16), printed p. 18]: an equality of expected utilities, not of expected spread capture and cost. Its consequences for the number of market makers are in mod:grossman-miller.

Per-share fees

Let a fee η, in currency per unit, be paid by each side per share traded. At date 2 a position is unwound either then or at date 3, so the fee enters both sides of each comparison and cancels:

$$q_2^j = \frac{\mathbb{E}[S_3 - \eta \mid \epsilon_2] - (S_2 - \eta)}{\gamma\sigma^2} \; (j \in \{MM, LT1\}), \qquad q_2^{LT2} = \frac{\mathbb{E}[S_3 + \eta \mid \epsilon_2] - (S_2 + \eta)}{\gamma\sigma^2},$$

and clearing again gives $S_2 = \mu + \epsilon_2$. At date 1, for $i > 0$, LT1 compares selling now at $S_1 - \eta$ with selling at date 2 at $S_2 - \eta$, so its fee cancels; a market maker buys now at $S_1 + \eta$ and expects to sell at $S_2 - \eta$, a round trip:

$$q_1^{LT1} = \frac{\mathbb{E}[S_2 - \eta] - (S_1 - \eta)}{\gamma\sigma^2} = \frac{\mu - S_1}{\gamma\sigma^2}, \qquad q_1^{MM} = \frac{\mathbb{E}[S_2 - \eta] - (S_1 + \eta)}{\gamma\sigma^2} = \frac{\mu - S_1 - 2\eta}{\gamma\sigma^2}.$$

Clearing, $i = nq_1^{MM} + q_1^{LT1}$, gives $(n+1)(\mu - S_1) - 2n\eta = \gamma\sigma^2 i$, so

$$S_1 = \mu - \gamma\sigma^2 \frac{i}{n+1} - \frac{2n}{n+1}\eta, \qquad q_1^{LT1} = \frac{i}{n+1} + \frac{2n}{n+1}\frac{\eta}{\gamma\sigma^2}.$$

The fee deepens the discount and LT1 keeps more inventory than without fees: it pays its own fee directly and most of the market makers' round-trip fee through a lower price. If $|i|$ is small relative to η the desired trade can fall into a no-trade region. With $\eta = 0$ the date-1 results above return.

Liquidity measures

Price-impact slope. Without fees, LT1's date-1 sale is $Q_1^{LT1} = i - q_1^{LT1} = \frac{n}{n+1} i$, so $\frac{i}{n+1} = \frac{Q_1^{LT1}}{n}$ and

$$S_1 = \mu - \frac{\gamma\sigma^2}{n} Q_1^{LT1} = \mu + \lambda Q_1^{LT1}, \qquad \lambda = -\frac{\gamma\sigma^2}{n},$$

with Q_1^{LT1} the quantity sold. A more liquid market has a smaller $|\lambda|$: more market makers, lower risk aversion or lower interim volatility.

Return reversal. Add a date 0 and public news ϵ_1 before LT1 trades, with efficient prices $\mu_0 = \mathbb{E}[S_3]$, $\mu_1 = \mathbb{E}[S_3 \mid \epsilon_1]$, $\mu_2 = \mathbb{E}[S_3 \mid \epsilon_1, \epsilon_2]$. With $S_0 = \mu_0$, $S_1 = \mu_1 + \lambda Q_1^{LT1}$ and $S_2 = \mu_2$,

$$\Delta_1 = S_1 - S_0 = \epsilon_1 + \lambda Q_1^{LT1}, \qquad \Delta_2 = S_2 - S_1 = \epsilon_2 - \lambda Q_1^{LT1}.$$

If liquidity demand is independent of the news, all cross terms vanish except the order-pressure term:

$$\operatorname{Cov}(\Delta_1, \Delta_2) = \operatorname{Cov}\left(\epsilon_1 + \lambda Q_1^{LT1}, \epsilon_2 - \lambda Q_1^{LT1}\right) = -\lambda^2 \operatorname{Var}(Q_1^{LT1}) < 0.$$

The first move contains order pressure and the second reverses it; as $|\lambda| \to 0$ the price process tends to the efficient-price martingale.

Limitations

The only friction is inventory carried between dates 1 and 2, and LT2's arrival is certain, so the derivation says nothing about uncertain arrival times; fees are per share and equal for every side. The free-entry condition of the original equates expected utilities and is not solved here. The reversal result needs liquidity demand independent of news. The entries' Pitfalls fields discuss the remaining limits.

1.1.2 Kyle single-auction model

Purpose. Derive the unique linear equilibrium of Kyle's one-period auction: the insider's trading intensity, the price-impact slope, the insider's profit and the information the price reveals. Compendium entry: mod:kyle-single-auction.

Sources. [catalogue-0496, chap. "Strategic Trader Models I: Informed Traders", pp. 91–98, eqs. (4.3)–(4.5), (4.17), (4.20)]; the setting as in [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", pp. 31–32], whose printed λ on p. 33 is the reciprocal (corrected in its errata).

Implementation and checks. `trading_notes.numerical.market_making_models.microstructure_models.KyleAuction`; `trading_notes.numerical.market_making_models.lob_model_derivations_checks` part 3 (model index).

Assumptions

i. The asset is worth $v \sim N(\mu, \sigma_v^2)$.
ii. One risk-neutral insider observes v and submits $x(v)$.
iii. Price-insensitive noise traders submit $u \sim N(0, \sigma_u^2)$, independent of v; the insider does not see u.
iv. Competitive risk-neutral market makers see only $y = x(v) + u$ and set $S(y) = \mathbb{E}[v \mid y]$, earning zero expected profit given y.
v. One auction, after which v is realised. Equilibrium: the insider best-responds to the price rule, and the price rule is Bayes-consistent with the insider's strategy.

Variables. v and S are in price units, orders x, u and y in shares, so λ is in price per share and β in shares per unit of price. The insider's information set is $\{v\}$, the market makers' $\{y\}$. The insider's profit is $\Pi = x(v)\big(v - S(x(v) + u)\big)$. Conjecture a linear equilibrium

$$S(y) = \mu + \lambda y, \qquad x(v) = \beta(v - \mu).$$

The insider's best response

Given the price rule and $\mathbb{E}[u] = 0$,

$$\mathbb{E}_u\big[x\big(v - \mu - \lambda(x + u)\big) \mid v\big] = x(v - \mu) - \lambda x^2,$$

strictly concave in x for $\lambda > 0$. The first-order condition $v - \mu - 2\lambda x = 0$ gives

$$x^\star(v) = \frac{v - \mu}{2\lambda}, \qquad \beta = \frac{1}{2\lambda}.$$

The market makers' pricing

Under $x(v) = \beta(v-\mu)$, $y = \beta(v-\mu)+u$, and (v, y) is jointly normal, so the conditional expectation is the linear projection

$$\mathbb{E}[v \mid y] = \mu + \frac{\mathrm{Cov}(v,y)}{\mathrm{Var}(y)}\, y = \mu + \frac{\beta\sigma_v^2}{\beta^2\sigma_v^2+\sigma_u^2}\, y, \qquad \text{so} \qquad \lambda = \frac{\beta\sigma_v^2}{\beta^2\sigma_v^2+\sigma_u^2}.$$

The fixed point

Substitute $\beta = 1/(2\lambda)$: $\lambda(\beta^2\sigma_v^2+\sigma_u^2) = \beta\sigma_v^2$ becomes $\frac{\sigma_v^2}{4\lambda} + \lambda\sigma_u^2 = \frac{\sigma_v^2}{2\lambda}$, so $\lambda^2\sigma_u^2 = \frac{\sigma_v^2}{4}$ and, taking the positive root,

$$\lambda^\star = \frac{\sigma_v}{2\sigma_u}, \qquad \beta^\star = \frac{\sigma_u}{\sigma_v}, \qquad x^\star(v) = \frac{\sigma_u}{\sigma_v}(v-\mu), \qquad S^\star(y) = \mu + \frac{\sigma_v}{2\sigma_u}\, y.$$

There is no other linear equilibrium; nonlinear equilibria are not excluded [catalogue-0496, p. 97].

Profit, efficiency and revealed information

Conditional on v, $\mathbb{E}_u[\Pi \mid v] = x^\star(v)(v-\mu) - \lambda^\star x^\star(v)^2 = \frac{(v-\mu)^2}{2\lambda^\star} - \frac{(v-\mu)^2}{4\lambda^\star} = \frac{(v-\mu)^2}{4\lambda^\star}$, and ex ante

$$\mathbb{E}[\Pi] = \frac{\sigma_v^2}{4\lambda^\star} = \frac{\sigma_v\sigma_u}{2}.$$

Market makers break even given y, $\mathbb{E}[v - S^\star(y) \mid y] = 0$, so the price is semi-strong efficient for their information. In equilibrium $\beta^{\star 2}\sigma_v^2 = \sigma_u^2$, so $\mathrm{Var}(y) = 2\sigma_u^2$ and $\mathrm{Cov}(v,y) = \sigma_u\sigma_v$:

$$\mathrm{Var}(v \mid y) = \sigma_v^2 - \frac{\mathrm{Cov}(v,y)^2}{\mathrm{Var}(y)} = \sigma_v^2 - \frac{\sigma_u^2\sigma_v^2}{2\sigma_u^2} = \tfrac{1}{2}\sigma_v^2,$$

whatever the order flow: one auction reveals half of the insider's information.

Interpretation and limiting cases. The insider trades only part of the mispricing because a larger order reveals more through y; higher σ_u lowers $\lambda^\star$ and raises the insider's order and profit, higher σ_v raises $\lambda^\star$. The impact is informational and permanent, unlike the Grossman–Miller inventory pressure, which reverses.

Limitations

The equilibrium is unique only among linear ones, and it relies on normality of v and u, a single auction and an insider who does not see u. The entry's Pitfalls field discusses the linear price rule's limits.

1.1.3 Static limit-order depth

Purpose. Derive the depth at which a risk-neutral market maker posts each limit order when fills decay exponentially with depth. Compendium entry: mod:static-limit-order-depth.

Source. [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", pp. 28–30, eq. (2.3)].

Implementation and checks. `trading_notes.numerical.market_making_models.lob_model_derivations_checks` part 4, a grid search with no domain class (model index).

Assumptions

A risk-neutral market maker posts a sell order at $S_t + \delta^+$ and a buy order at $S_t - \delta^-$. A buy (sell) market order arrives with probability p^+ (p^-) and reaches depth δ with probability $e^{-\kappa^\pm\delta}$; there is no adverse selection, inventory cost or reaction by other traders. Depths are in price units and $\kappa^\pm > 0$ in inverse price units; $p^\pm$ are probabilities, not prices. The decision is made once, before the market order arrives.

Derivation

Expected profit per decision is

$$\mathbb{E}[\Pi(\delta^+,\delta^-)] = p^+ e^{-\kappa^+\delta^+}\delta^+ + p^- e^{-\kappa^-\delta^-}\delta^-,$$

which separates by side. On each side, $\frac{d}{d\delta}\left(p\, e^{-\kappa\delta}\delta\right) = p\, e^{-\kappa\delta}(1-\kappa\delta)$, which is positive for $\delta < 1/\kappa$ and negative beyond (own check: the second derivative at $\delta = 1/\kappa$ is $-p\kappa e^{-1} < 0$), so

$$\delta^{\pm,\star} = \frac{1}{\kappa^\pm},$$

the mean depth reached by market orders on that side.

Result and interpretation

Each side's optimal depth depends only on its own fill-decay rate $\kappa^\pm$. The arrival probabilities scale profit but not the optimal depth.

Limitations

Without adverse selection, inventory cost or reaction by other traders, the depth cannot respond to position or to informed flow; the entry's Pitfalls field lists the source's own reservations.

1.1.4 Roll's serial-covariance spread estimator

Purpose. Derive the variance and autocovariances of trade-price changes in Roll's model, and from them the spread estimator. Compendium entry: res:roll-spread-estimator, in the chapter's *Market organisation, informed trading and liquidity measurement* pathway.

Sources. [alpha-harris, chap. "Volatility", box, p. 415]; [s6-hasbrouck, sec. 3.4, pp. 29–30].

Implementation and checks. `trading_notes.numerical.market_making_models.microstructure_models.RollModel`; `trading_notes.numerical.market_making_models.lob_model_derivations_checks` part 5 (model index).

Assumptions

Let $P_t = V_t + \frac{1}{2} S Q_t$ with $V_t = V_{t-1} + \varepsilon_t$, $\mathbb{E}\varepsilon_t = 0$, $\operatorname{Var} \varepsilon_t = \sigma^2$, and $Q_t = \pm 1$ with probability one-half each, independent of earlier signs and of the innovations. S is the full spread, in price units, and t counts trades, so price changes are between successive trades. Then

$$\Delta P_t = \varepsilon_t + \tfrac{1}{2} S (Q_t - Q_{t-1}).$$

Derivation

Variance. ε_t is independent of the sign terms, and $\mathbb{E}[(Q_t - Q_{t-1})^2] = \mathbb{E}Q_t^2 - 2\mathbb{E}[Q_t Q_{t-1}] + \mathbb{E}Q_{t-1}^2 = 1 - 0 + 1 = 2$, so

$$\operatorname{Var}(\Delta P_t) = \sigma^2 + \tfrac{1}{4} S^2 \cdot 2 = \sigma^2 + \tfrac{1}{2} S^2.$$

First autocovariance. The innovations are serially independent and independent of the signs, so only the sign terms contribute:

$$\operatorname{Cov}(\Delta P_t, \Delta P_{t-1}) = \tfrac{1}{4} S^2 \, \mathbb{E}\big[(Q_t - Q_{t-1})(Q_{t-1} - Q_{t-2})\big] = \tfrac{1}{4} S^2 \big(\mathbb{E}[Q_t Q_{t-1}] - \mathbb{E}[Q_t Q_{t-2}] - \mathbb{E}Q_{t-1}^2 + \mathbb{E}[Q_{t-1} Q_{t-2}]\big) = -\tfrac{1}{4} S^2,$$

since every cross-product has zero expectation and $\mathbb{E}Q_{t-1}^2 = 1$. Autocovariances of order two or more are zero, because ΔP_t and ΔP_{t-k} share no sign or innovation for $k \geq 2$.

Result

Hence, when the first autocovariance is negative,

$$S = 2\sqrt{-\operatorname{Cov}(\Delta P_t, \Delta P_{t-1})}.$$

Limitations

The estimator is defined only when the first autocovariance is negative. How correlated signs and informed trading bias the estimate is in the entry's Pitfalls.

Sources outside the catalogue

grossman-miller-1988-nber Sanford J. Grossman and Merton H. Miller, “Liquidity and Market Structure”, NBER Working Paper 2641, July 1988; the working-paper version of grossman-miller-1988, whose pages and numbering may differ.

Every other cited key links to its entry in the pinned reading-list catalogue in the guide copy under guide/.